

MUGUR-AKSY, Russian FederationWMO: **362780**Lat: **50.382N**Lon: **90.434E**Elev: **1850**StdP: **80.99**Time Zone: **7.00 (E07)**Period: **90-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.1	(c) -29.5	(d) -35.5	(e) 0.2	(f) -32.0	(g) -32.7	(h) 0.2	(i) -28.8	(j) 8.5	(k) -14.5	(l) 7.9	(m) -14.7	(n) 1.4	(o) 280	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 11.7	(c) 24.8	(d) 12.0	(e) 22.9	(f) 11.6	(g) 21.3	(h) 11.1	(i) 13.2	(j) 21.3	(k) 12.5	(l) 20.5	(m) 11.9	(n) 19.5	(o) 3.7	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
10.4	9.8	15.0	9.3	9.2	15.1	8.4	8.6	14.9	43.2	21.3	41.1	20.7	39.4	19.6	16.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.4	(b) 8.2	(c) 7.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 9.4	(b) 8.2	(c) 7.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.4	-20.0	-16.3	-6.6	0.7	6.4	13.5	14.7	12.6	6.3	-1.2	-10.6	-17.3		
	DBStd	12.86	4.71	5.54	5.22	4.64	4.42	3.22	2.41	2.62	4.39	5.01	5.62	5.07		
	HDD10.0	4542	931	738	514	278	128	9	1	8	123	347	619	847		
	HDD18.3	7209	1189	971	772	528	371	148	114	176	360	604	869	1106		
	CDD10.0	378	0	0	0	0	16	113	147	89	13	0	0	0		
	CDD18.3	4	0	0	0	0	0	2	2	0	0	0	0	0		
	CDH23.3	117	0	0	0	0	1	46	59	10	0	0	0	0		
	CDH26.7	6	0	0	0	0	0	2	4	0	0	0	0	0		
(14) Wind	WSAvg	2.9	1.9	2.6	2.9	3.3	3.7	2.9	3.1	2.7	2.9	2.9	3.1	2.4		
(15) Precipitation	PrecAvg	134	2	2	4	4	8	22	40	28	9	5	6	3		
	PrecMax	323	10	13	24	12	19	51	157	123	44	13	25	11		
	PrecMin	59	0	0	0	0	1	3	9	3	0	0	0	0		
	PrecStd	56	2	3	5	3	5	14	28	25	9	4	6	3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-6.1	0.8	8.0	15.9	22.6	26.5	26.9	24.8	21.3	14.0	5.2	-2.9		
		MCWB	-8.1	-2.8	1.6	7.2	10.1	12.4	12.6	11.7	9.9	5.7	1.6	-4.6		
	2%	DB	-8.5	-2.6	6.0	12.7	19.5	24.8	25.2	22.4	18.7	11.7	2.1	-5.3		
		MCWB	-10.2	-5.3	0.2	4.9	8.8	11.8	12.5	11.5	9.0	4.4	-0.7	-7.1		
	5%	DB	-10.8	-5.2	4.0	10.3	17.2	22.8	23.1	20.8	16.6	9.5	-0.9	-7.7		
		MCWB	-12.2	-7.5	-0.8	3.4	7.4	11.4	11.9	11.1	8.1	3.2	-3.3	-9.2		
	10%	DB	-12.8	-7.5	2.0	8.5	15.0	20.9	21.2	19.2	14.3	6.8	-2.9	-9.5		
		MCWB	-14.0	-9.4	-2.0	2.3	6.6	10.6	11.5	10.6	7.5	1.7	-5.2	-10.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.0	-2.4	2.6	8.3	11.1	14.0	14.3	13.9	11.0	6.2	2.3	-4.3		
		MCDB	-6.2	0.4	6.7	14.5	20.7	21.8	21.9	19.5	19.1	13.0	4.9	-3.2		
	2%	WB	-10.1	-5.3	0.6	5.7	9.4	12.7	13.3	12.7	9.7	4.8	-0.6	-7.0		
		MCDB	-8.5	-2.9	5.3	12.2	18.4	21.3	22.0	19.5	17.3	10.7	2.0	-5.3		
	5%	WB	-12.2	-7.4	-0.5	4.0	8.1	12.1	12.7	11.9	8.7	3.3	-3.3	-9.1		
		MCDB	-10.9	-5.2	3.6	9.2	16.2	20.5	21.2	18.9	15.8	9.1	-1.1	-7.7		
	10%	WB	-14.0	-9.4	-1.9	2.7	6.8	11.3	12.1	11.1	7.6	1.8	-5.0	-10.7		
		MCDB	-12.8	-7.4	1.6	7.8	14.2	19.7	19.8	17.9	14.1	6.6	-3.0	-9.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	11.6	12.0	11.4	12.3	12.6	11.7	11.4	12.2	10.6	9.8	9.1		
		MCDBR	11.0	13.2	13.8	14.1	15.8	15.7	15.0	14.4	15.4	14.4	11.9	8.9		
	5% WB	MCWBR	9.7	10.7	9.2	8.5	8.1	6.9	6.3	6.3	7.9	8.8	9.3	7.7		
		MCWBR	11.2	12.8	12.5	12.8	15.2	14.0	12.7	12.3	14.1	13.6	11.8	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.224	0.248	0.295	0.342	0.340	0.341	0.341	0.323	0.287	0.273	0.247	0.224		
	taud		2.299	2.255	2.195	2.240	2.308	2.399	2.425	2.447	2.495	2.448	2.351	2.288		
	Ebn at Noon		846	904	908	899	915	915	909	912	917	868	803	784		
	Edh at Noon		64	89	115	125	122	113	108	100	85	73	61	56		
(44) All-Sky Solar Radiation	RadAvg		1.37	2.37	3.92	5.32	6.27	6.32	5.98	5.33	4.22	2.65	1.40	1.04		
	RadStd		0.06	0.08	0.17	0.18	0.39	0.34	0.36	0.31	0.24	0.19	0.09	0.05		

Historical Trends

		Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	DBAvg	N/A	N/A	-2.37	N/A	-0.76	-1.11	N/A	N/A

Nomenclature: See separate page